

Lab 4 – Big Data Processing using EMR + Spark + Athena + Step Functions

Summary of the Lab

In this lab, we:

- Process rental marketplace data using **Spark on EMR**
- Store transformed data in **S3 (Data Lake)**
- Use **Glue Crawlers** to infer schema
- Query using **Athena**
- Automate workflow using **Step Functions**
- Execute same jobs using **EMR Serverless**

Prerequisites

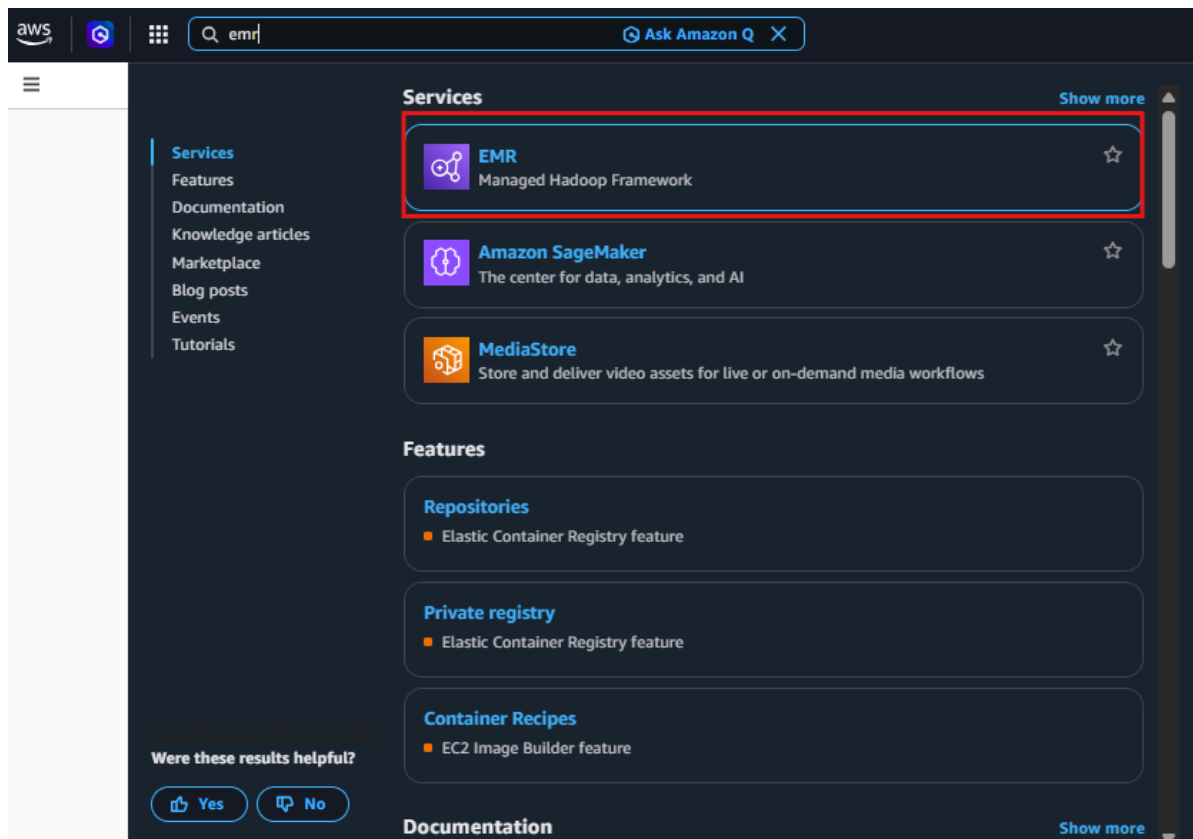
- S3 bucket created
- Raw datasets uploaded:
 - locations.csv
 - rental_transactions.csv
 - users.csv
 - vehicles.csv
- Spark scripts ready:
 - spark_agg_1.py
 - spark_agg_2.py

Part 1 – Test Spark Jobs Locally

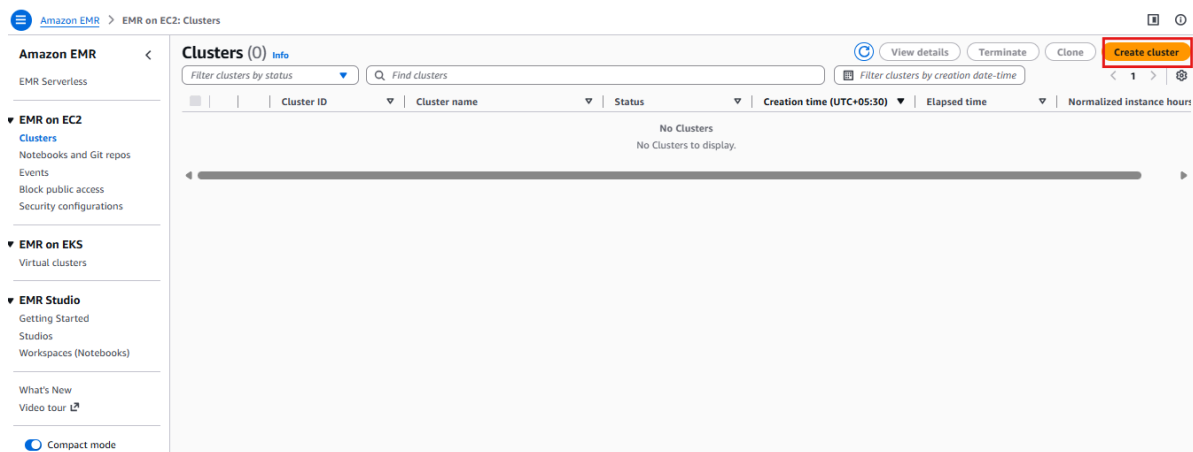
- Open terminal
- Navigate to Spark project directory
- Ensure Docker engine is running
- Run first Spark job
- Verify:
 - Transaction-level metrics
 - User-level metrics
- Run second Spark job
- Verify:
 - Location-level metrics
 - Vehicle-type metrics
- Ensure no syntax or logic errors

Part 2 – Create EMR Cluster (Provisioned)

- Go to AWS Console → EMR



- Click **Create Cluster**



Cluster Configuration:

- Name: Spark Job Cluster

Amazon EMR > EMR on EC2: Clusters > Create cluster

Create cluster Info

Name your cluster and choose the applications that you want to install to your cluster.

Name and applications - required Info

Name
spark-job-cluster

Amazon EMR release Info
A release contains a set of applications which can be installed on your cluster.
emr-7.12.0

Application bundle

Spark Interactive 	Core Hadoop 	Flink 	HBase 	Presto 	Trino 	Custom
-----------------------	-----------------	-----------	-----------	------------	-----------	------------

☐ AmazonCloudWatchAgent
☐ 1.300032.2
☐ HCatalog 3.1.3
☐ Hue 4.11.0
☐ Livy 0.8.0
☐ Pig 0.17.0
☐ TensorFlow 2.19.0
☐ Zeppelin 0.11.1

☐ Flink 1.20.0
☐ Hadoop 3.4.1
☐ JupyterEnterpriseGateway 2.6.0
☐ Oozie 5.2.1
☐ Presto 0.287
☐ Tez 0.10.2
☐ ZooKeeper 3.9.3

☐ HBase 2.6.2
☐ Hive 3.1.3
☐ JupyterHub 1.5.0
☐ Phoenix 5.2.1
☒ Spark 3.5.6
☐ Trino 476

Summary Info

Name and applications - required

Name
spark-job-cluster

Amazon EMR release
emr-7.12.0

Application bundle
Custom (Spark 3.5.6)

Cluster configuration - required

Uniform instance groups
Primary (r8g.xlarge), Core (r8g.xlarge), Task (r8g.xlarge)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance
Task size: 1 instance

- EMR Version: 7.1.0

Amazon EMR > EMR on EC2: Clusters > Create cluster

Create cluster Info

Name your cluster and choose the applications that you want to install to your cluster.

Name and applications - required Info

Name
spark-job-cluster

Amazon EMR release Info
A release contains a set of applications which can be installed on your cluster.
emr-7.10.0

Application bundle

Spark Interactive 	Core Hadoop 	Flink 	HBase 	Presto 	Trino 	Custom
-----------------------	-----------------	-----------	-----------	------------	-----------	------------

☐ AmazonCloudWatchAgent
☐ 1.300032.2
☐ HCatalog 3.1.3
☐ Hue 4.11.0
☐ Livy 0.8.0
☐ Pig 0.17.0
☐ TensorFlow 2.18.0
☐ Zeppelin 0.11.1

☐ Flink 1.20.0
☐ Hadoop 3.4.1
☐ JupyterEnterpriseGateway 2.6.0
☐ Oozie 5.2.1
☐ Presto 0.287
☐ Tez 0.10.2
☐ ZooKeeper 3.9.3

☐ HBase 2.6.2
☐ Hive 3.1.3
☐ JupyterHub 1.5.0
☐ Phoenix 5.2.1
☒ Spark 3.5.5
☐ Trino 470

Summary Info

Name and applications - required

Name
spark-job-cluster

Amazon EMR release
emr-7.10.0

Application bundle
Custom (Spark 3.5.5)

Cluster configuration - required

Uniform instance groups
Primary (r8g.xlarge), Core (r8g.xlarge), Task (r8g.xlarge)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance
Task size: 1 instance

- Application bundle —> Select only Spark

Amazon EMR > EMR on EC2: Clusters > Create cluster

Create cluster Info

Name your cluster and choose the applications that you want to install to your cluster.

Name and applications - required Info

Name
spark-job-cluster

Amazon EMR release Info
A release contains a set of applications which can be installed on your cluster.
emr-7.10.0

Application bundle

Spark Interactive 	Core Hadoop 	Flink 	HBase 	Presto 	Trino 	Custom
-----------------------	-----------------	-----------	-----------	------------	-----------	------------

☐ AmazonCloudWatchAgent
☐ 1.300032.2
☐ HCatalog 3.1.3
☐ Hue 4.11.0
☐ Livy 0.8.0
☐ Pig 0.17.0
☐ TensorFlow 2.18.0
☐ Zeppelin 0.11.1

☐ Flink 1.20.0
☐ Hadoop 3.4.1
☐ JupyterEnterpriseGateway 2.6.0
☐ Oozie 5.2.1
☐ Presto 0.287
☐ Tez 0.10.2
☐ ZooKeeper 3.9.3

☐ HBase 2.6.2
☐ Hive 3.1.3
☐ JupyterHub 1.5.0
☐ Phoenix 5.2.1
☒ Spark 3.5.5
☐ Trino 470

Summary Info

Name and applications - required

Name
spark-job-cluster

Amazon EMR release
emr-7.10.0

Application bundle
Custom (Spark 3.5.5)

Cluster configuration - required

Uniform instance groups
Primary (r8g.xlarge), Core (r8g.xlarge), Task (r8g.xlarge)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance
Task size: 1 instance

- Keep everything else default
- Under Identity and Access management (IAM) Roles

Amazon EMR > EMR on EC2: Clusters > Create cluster

▼ Identity and Access Management (IAM) roles - required [Info](#)

Choose or create a service role and instance profile for the EC2 instances in your cluster.

Amazon EMR service role [Info](#)

The service role is an IAM role that Amazon EMR assumes to provision resources and perform service-level actions with other AWS services.

☒ **Choose an existing service role**
 Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ **Create a service role**
 Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role

[Refresh](#)

EC2 instance profile for Amazon EMR

The instance profile assigns a role to every EC2 instance in a cluster. The instance profile must specify a role that can access the resources for your steps and bootstrap actions.

☒ **Choose an existing instance profile**
 Select a default role or a custom instance profile with IAM policies attached so that your cluster can interact with your resources in Amazon S3.

☐ **Create an instance profile**
 Let Amazon EMR create a new instance profile so that you can specify a custom set of resources for it to access in Amazon S3.

Instance profile

[Refresh](#)

Summary [Info](#)

Name and applications - required

Name
spark-job-cluster

Amazon EMR release
emr-7.10.0

Application bundle
Custom (Spark 3.5.5)

Cluster configuration - required

Uniform instance groups
Primary (r8g.xlarge), Core (r8g.xlarge), Task (r8g.xlarge)

Cluster scaling and provisioning - required

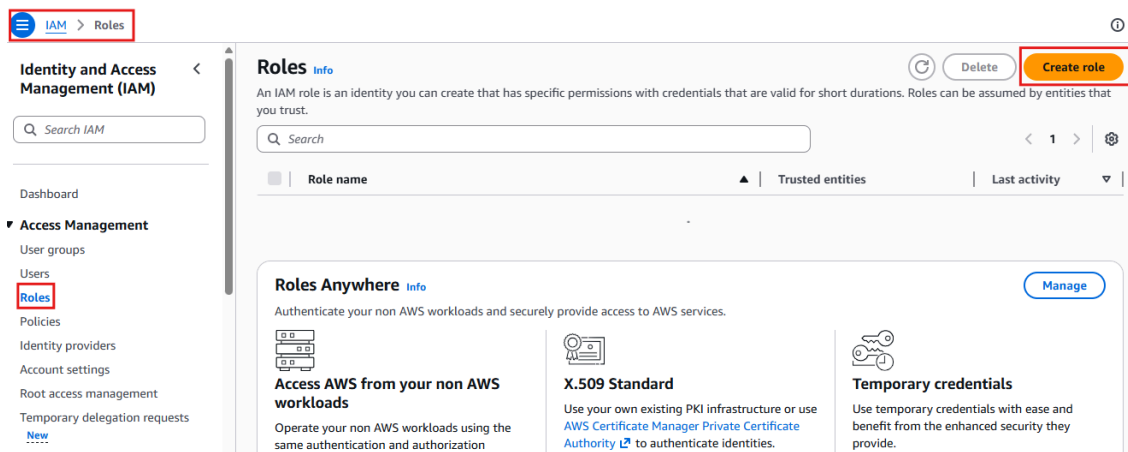
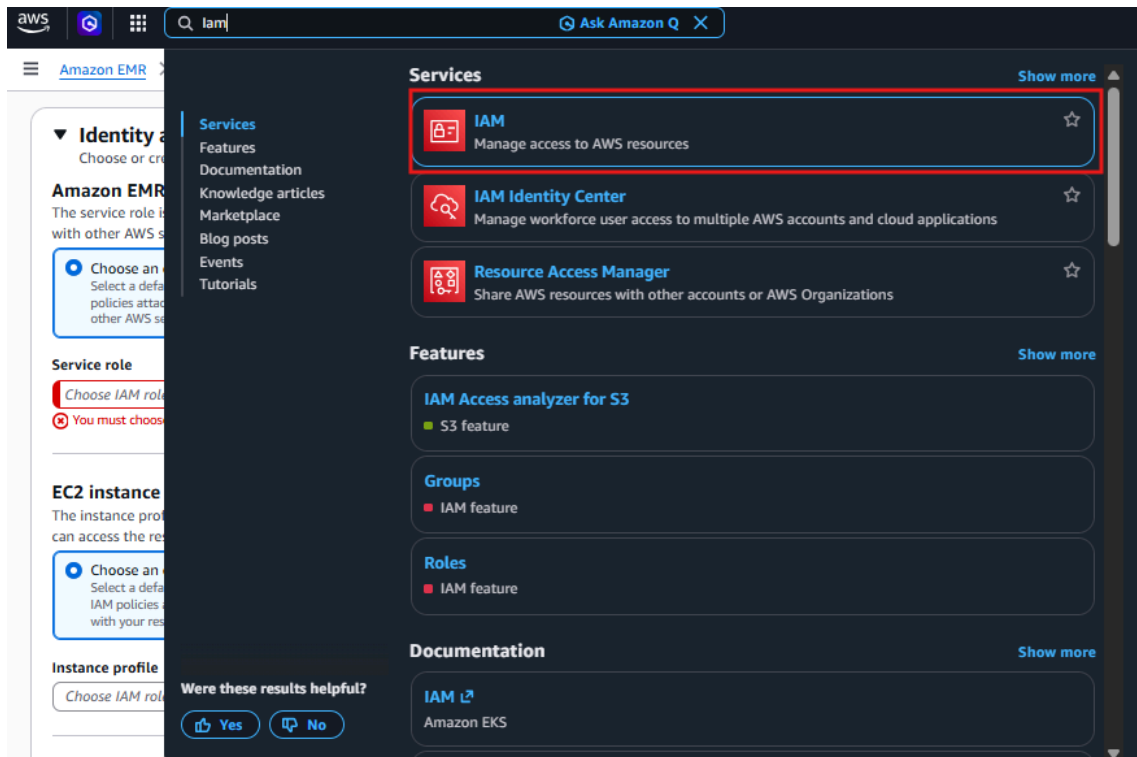
Provisioning configuration
Core size: 1 instance
Task size: 1 instance

Configure IAM roles
You must choose a service role and instance

Create IAM Roles

1. EMR Service Role

- IAM → Create Role



- Select service: EMR

IAM > Roles > Create role ⓘ

perform actions in this account. account.

☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EMR ▼

Choose a use case for the specified service.

Use case

- ☒ EMR
Allows Elastic MapReduce to call AWS services such as EC2 on your behalf.
- ☐ EMR Role for EC2
Allows EC2 instances in an Elastic MapReduce cluster to call AWS services such as S3 on your behalf.
- ☐ EMR - Cleanup
Allows EMR to terminate instances and delete resources from EC2 on your behalf.

- Choose use case: EMR

IAM > Roles > Create role ⓘ

perform actions in this account. account.

☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EMR ▼

Choose a use case for the specified service.

Use case

- ☒ EMR
Allows Elastic MapReduce to call AWS services such as EC2 on your behalf.
- ☐ EMR Role for EC2
Allows EC2 instances in an Elastic MapReduce cluster to call AWS services such as S3 on your behalf.
- ☐ EMR - Cleanup
Allows EMR to terminate instances and delete resources from EC2 on your behalf.

- Click next
- Name it: EMR Custom Service Role

IAM > Roles > Create role ⓘ

Step 1
● Select trusted entity

Step 2
● Add permissions

Step 3
● **Name, review, and create**

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.
emr-custom-service-role
Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

Description
Add a short explanation for this role.
Allows Elastic MapReduce to call AWS services such as EC2 on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/\[\]\#%*^!~:~"

- Click create role

Step 2: Add permissions Edit

Permissions policy summary

Policy name	Type	Attached as
AmazonEMRServicePolicy_v2	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional [Info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create role](#)

2. EMR EC2 Role

- Navigate to IAM → Click Create Role

Roles (988) [Info](#) [Refresh](#) [Delete](#) [Create role](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

☐	Role name	Trusted entities	Last activity
☐	academysqltos3role	AWS Service: rds	-
☐	academysqltos3role12	AWS Service: rds	547 days ago
☐	AffRole2	AWS Service: ec2	572 days ago
☐	airseriesrole2	AWS Service: rds	480 days ago
☐	AmazonBedrockExecutionRoleForAgents_EMKVOWN8DB7	AWS Service: bedrock	472 days ago
☐	AmazonBedrockExecutionRoleForAgents_FGW8ET8571I	AWS Service: bedrock	471 days ago
☐	AmazonBedrockExecutionRoleForAgents_OKJFIV2KZW	AWS Service: bedrock	-
☐	AmazonBedrockExecutionRoleForKnowledgeBase_6jxtt	AWS Service: bedrock	289 days ago

- Select service: EMR

- Go back to EMR
- Refresh roles

Amazon EMR > EMR on EC2: Clusters > Create cluster

▼ Identity and Access Management (IAM) roles - required [Info](#)

Choose or create a service role and instance profile for the EC2 instances in your cluster.

Amazon EMR service role [Info](#)

The service role is an IAM role that Amazon EMR assumes to provision resources and perform service-level actions with other AWS services.

☒ Choose an existing service role
Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ Create a service role
Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role

Choose IAM role

You must choose a role to use as a service role.

EC2 instance profile for Amazon EMR

The instance profile assigns a role to every EC2 instance in a cluster. The instance profile must specify a role that can access the resources for your steps and bootstrap actions.

☒ Choose an existing instance profile
Select a default role or a custom instance profile with IAM policies attached so that your cluster can interact with your resources in Amazon S3.

☐ Create an instance profile
Let Amazon EMR create a new instance profile so that you can specify a custom set of resources for it to access in Amazon S3.

Instance profile

Choose IAM role

Custom automatic scaling role - optional

When a custom automatic scaling rule triggers, Amazon EMR assumes this role to add and terminate EC2 instances. [Learn more](#)

Custom automatic scaling role

Choose IAM role

Create IAM role

Summary [Info](#)

Name and applications - required

Name
spark-job-cluster

Amazon EMR release
emr-7.10.0

Application bundle
Custom (Spark 3.5.5)

Cluster configuration - required

Uniform instance groups
Primary (r8g.xlarge), Core (r8g.xlarge), Task (r8g.xlarge)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance
Task size: 1 instance

Configure IAM roles
You must choose a service role and instance profile before you create this cluster.

Choose IAM roles

- Select created roles

Amazon EMR > EMR on EC2: Clusters > Create cluster

with other AWS services.

☒ Choose an existing service role
Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ Create a service role
Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role

emr-custom-service-role

EC2 instance profile for Amazon EMR

The instance profile assigns a role to every EC2 instance in a cluster. The instance profile must specify a role that can access the resources for your steps and bootstrap actions.

☒ Choose an existing instance profile
Select a default role or a custom instance profile with IAM policies attached so that your cluster can interact with your resources in Amazon S3.

☐ Create an instance profile
Let Amazon EMR create a new instance profile so that you can specify a custom set of resources for it to access in Amazon S3.

Instance profile

emr-ec2-custom-role

Custom automatic scaling role - optional

When a custom automatic scaling rule triggers, Amazon EMR assumes this role to add and terminate EC2 instances. [Learn more](#)

Custom automatic scaling role

Choose IAM role

Create IAM role

Summary [Info](#)

Name and applications - required

Name
spark-job-cluster

Amazon EMR release
emr-7.10.0

Application bundle
Custom (Spark 3.5.5)

Cluster configuration - required

Uniform instance groups
Primary (r8g.xlarge), Core (r8g.xlarge), Task (r8g.xlarge)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance
Task size: 1 instance

Cancel

Create cluster

- Click **Create Cluster**

Amazon EMR > EMR on EC2: Clusters > Create cluster

with other AWS services.

☒ Choose an existing service role
Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☐ Create a service role
Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Service role

emr-custom-service-role

EC2 instance profile for Amazon EMR

The instance profile assigns a role to every EC2 instance in a cluster. The instance profile must specify a role that can access the resources for your steps and bootstrap actions.

☒ Choose an existing instance profile
Select a default role or a custom instance profile with IAM policies attached so that your cluster can interact with your resources in Amazon S3.

☐ Create an instance profile
Let Amazon EMR create a new instance profile so that you can specify a custom set of resources for it to access in Amazon S3.

Instance profile

emr-ec2-custom-role

Custom automatic scaling role - optional

When a custom automatic scaling rule triggers, Amazon EMR assumes this role to add and terminate EC2 instances.

[Learn more](#)

Custom automatic scaling role

Choose IAM role

Create IAM role

Summary

Name and applications - required

Name
spark-job-cluster

Amazon EMR release
emr-7.10.0

Application bundle
Custom (Spark 3.5.5)

Cluster configuration - required

Uniform instance groups
Primary (r8g.xlarge), Core (r8g.xlarge), Task (r8g.xlarge)

Cluster scaling and provisioning - required

Provisioning configuration
Core size: 1 instance
Task size: 1 instance

Cancel Create cluster

- Wait until status shows **Waiting**

Amazon EMR > EMR on EC2: Clusters

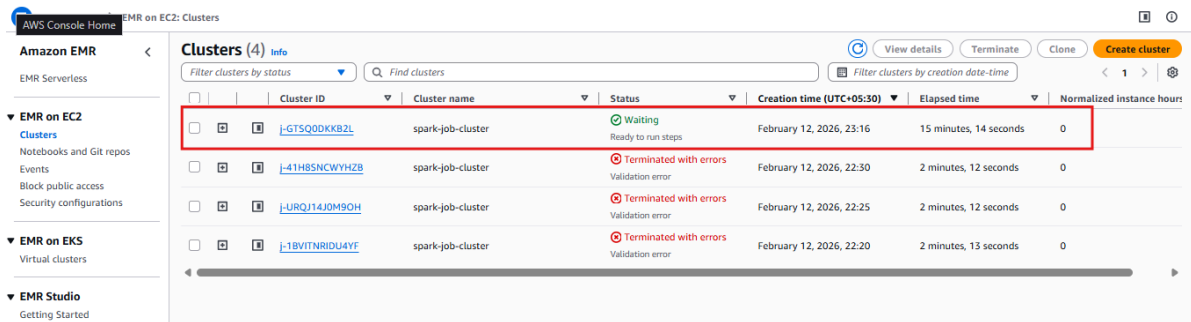
Clusters (4) Info

Filter clusters by status Find clusters Filter clusters by creation date-time

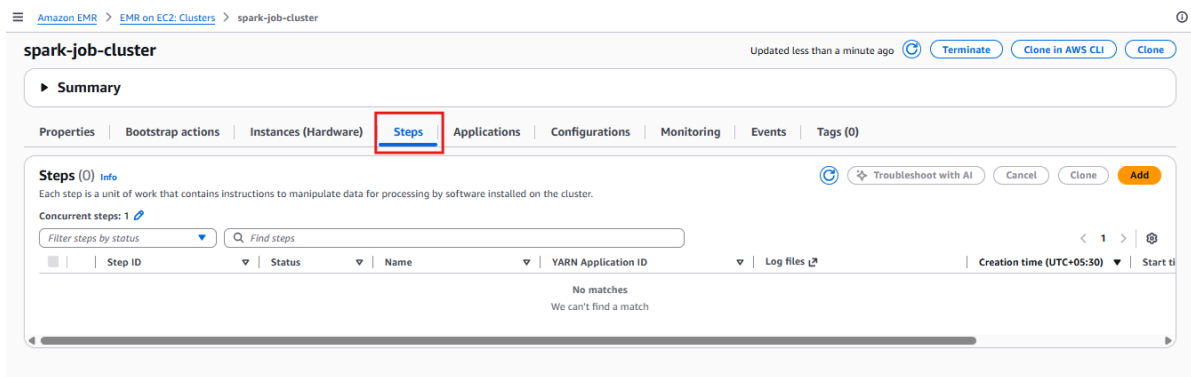
	Cluster ID	Cluster name	Status	Creation time (UTC+05:30)	Elapsed time	Normalized instance hours
☐	j-GTSQ0DKKB2L	spark-job-cluster	Waiting Ready to run steps	February 12, 2026, 23:16	15 minutes, 14 seconds	0
☐	j-41H8SNCWYH2B	spark-job-cluster	Terminated with errors Validation error	February 12, 2026, 22:30	2 minutes, 12 seconds	0
☐	j-URQJ14J0M9QH	spark-job-cluster	Terminated with errors Validation error	February 12, 2026, 22:25	2 minutes, 12 seconds	0
☐	j-1BVITNRIDU4YF	spark-job-cluster	Terminated with errors Validation error	February 12, 2026, 22:20	2 minutes, 13 seconds	0

Part 3 – Add Spark Jobs as Steps

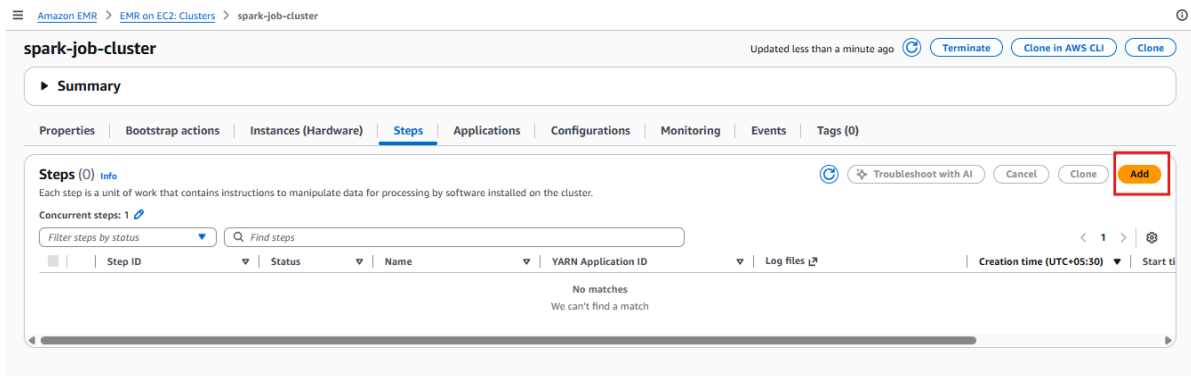
- Click on Cluster ID



- Go to **Steps**



- Click **Add Step**



Add Spark Job 1

- Select Type: Spark Application

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Add step Info

Step settings

Type

☐ Custom JAR
Adds a step that enables you to write a custom script to process your data using the Java programming language.

☒ **Spark application**
Adds a step that submits work to the Spark framework on the cluster.

☐ Streaming program
Adds a step that uses standard input to run mapper/reducer scripts and send results to standard output.

☐ Shell script
Troubleshoot your cluster.

Name

spark-job1

Deploy mode

☒ **Cluster mode**
Run your driver on a worker node.

☐ Client mode
Run your driver on the primary node as an external client.

Application location
Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

s3://bucket/prefix/object View i2 Browse S3

- Name: Spark Job 1

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Add step Info

Step settings

Type

☐ Custom JAR
Adds a step that enables you to write a custom script to process your data using the Java programming language.

☒ **Spark application**
Adds a step that submits work to the Spark framework on the cluster.

☐ Streaming program
Adds a step that uses standard input to run mapper/reducer scripts and send results to standard output.

☐ Shell script
Troubleshoot your cluster.

Name

spark-job1

Deploy mode

☒ **Cluster mode**
Run your driver on a worker node.

☐ Client mode
Run your driver on the primary node as an external client.

Application location
Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

s3://bucket/prefix/object View i2 Browse S3

- Application location: S3 path of spark_agg1.py

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Run your driver on the primary node as an external client.

Application location
Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

s3://ab-aws-de-labs/spark-scripts/spark-agg1.py View i2 Browse S3

Spark-submit options - optional
Specify other options for spark-submit.

--class, org.apache.spark.examples.SparkPi

Arguments - optional Info
Specify optional arguments for your script.

/usr/lib/spark/examples/jars/spark-examples.jar10

Step action

Action if step fails
The action to take when the step fails.

☒ **Continue**
Continues to the next step in the queue.

☐ Cancel and wait
Cancels any pending steps and returns the cluster to the waiting state.

☐ Terminate cluster
Shuts down the cluster.

Cancel Add step

- Action on failure: Continue

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Run your driver on the primary node as an external client.

Application location
Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

View Browse S3

Spark-submit options - optional
Specify other options for spark-submit.

Arguments - optional [Info](#)
Specify optional arguments for your script.

Step action
Action if step fails
The action to take when the step fails.

☒ **Continue**
Continues to the next step in the queue.

☐ **Cancel and wait**
Cancels any pending steps and returns the cluster to the waiting state.

☐ **Terminate cluster**
Shuts down the cluster.

Cancel Add step

- Click Add Step

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Run your driver on the primary node as an external client.

Application location
Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

View Browse S3

Spark-submit options - optional
Specify other options for spark-submit.

Arguments - optional [Info](#)
Specify optional arguments for your script.

Step action
Action if step fails
The action to take when the step fails.

☒ **Continue**
Continues to the next step in the queue.

☐ **Cancel and wait**
Cancels any pending steps and returns the cluster to the waiting state.

☐ **Terminate cluster**
Shuts down the cluster.

Cancel Add step

Add Spark Job 2

- Type: Spark Application

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Add step Info

Step settings

Type

☐ Custom JAR
 Adds a step that enables you to write a custom script to process your data using the Java programming language.

☒ **Spark application**
 Adds a step that submits work to the Spark framework on the cluster.

☐ Streaming program
 Adds a step that uses standard input to run mapper/reducer scripts and send results to standard output.

☐ Shell script
 Troubleshoot your cluster.

Name

Enter a name

Deploy mode

☒ **Cluster mode**
 Run your driver on a worker node.

☐ Client mode
 Run your driver on the primary node as an external client.

Application location

Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

Spark-submit options - optional

Specify other options for spark-submit.

- Name: Spark Job 2

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Add step Info

Step settings

Type

☐ Custom JAR
 Adds a step that enables you to write a custom script to process your data using the Java programming language.

☒ **Spark application**
 Adds a step that submits work to the Spark framework on the cluster.

☐ Streaming program
 Adds a step that uses standard input to run mapper/reducer scripts and send results to standard output.

☐ Shell script
 Troubleshoot your cluster.

Name

Enter a name

Deploy mode

☒ **Cluster mode**
 Run your driver on a worker node.

☐ Client mode
 Run your driver on the primary node as an external client.

Application location

Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

Spark-submit options - optional

Specify other options for spark-submit.

- Application location: S3 path of spark_agg_2.py

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Add step Info

Step settings

Type

☐ Custom JAR
 Adds a step that enables you to write a custom script to process your data using the Java programming language.

☒ **Spark application**
 Adds a step that submits work to the Spark framework on the cluster.

☐ Streaming program
 Adds a step that uses standard input to run mapper/reducer scripts and send results to standard output.

☐ Shell script
 Troubleshoot your cluster.

Name

Enter a name

Deploy mode

☒ **Cluster mode**
 Run your driver on a worker node.

☐ Client mode
 Run your driver on the primary node as an external client.

Application location

Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

Spark-submit options - optional

Specify other options for spark-submit.

- Action on failure: Continue

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Run your driver on the primary node as an external client.

Application location
Path to a JAR with your application and dependencies (client deploy mode only supports a local path).
s3://ab-aws-de-labs/spark-scripts/spark-agg2.py View [View](#) [Browse S3](#)

Spark-submit options - optional
Specify other options for spark-submit.
--class, org.apache.spark.examples.SparkPI

Arguments - optional [Info](#)
Specify optional arguments for your script.
/usr/lib/spark/examples/jars/spark-examples.jar10

Step action
Action if step fails
The action to take when the step fails.
☒ Continue
Continues to the next step in the queue.
☐ Cancel and wait
Cancels any pending steps and returns the cluster to the waiting state.
☐ Terminate cluster
Shuts down the cluster.

Cancel Add step

- Click Add Step

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster > Add step

Run your driver on the primary node as an external client.

Application location
Path to a JAR with your application and dependencies (client deploy mode only supports a local path).
s3://ab-aws-de-labs/spark-scripts/spark-agg2.py View [View](#) [Browse S3](#)

Spark-submit options - optional
Specify other options for spark-submit.
--class, org.apache.spark.examples.SparkPI

Arguments - optional [Info](#)
Specify optional arguments for your script.
/usr/lib/spark/examples/jars/spark-examples.jar10

Step action
Action if step fails
The action to take when the step fails.
☒ Continue
Continues to the next step in the queue.
☐ Cancel and wait
Cancels any pending steps and returns the cluster to the waiting state.
☐ Terminate cluster
Shuts down the cluster.

Cancel Add step

- Wait for execution to complete

Amazon EMR > EMR on EC2: Clusters > spark-job-cluster

spark-job-cluster Updated less than a minute ago [Terminate](#) [Clone in AWS CLI](#) [Clone](#)

Summary

Properties | Bootstrap actions | Instances (Hardware) | **Steps** | Applications | Configurations | Monitoring | Events | Tags (0)

Steps (4) [Info](#) [Troubleshoot with AI](#) [Cancel](#) [Clone](#) [Add](#)

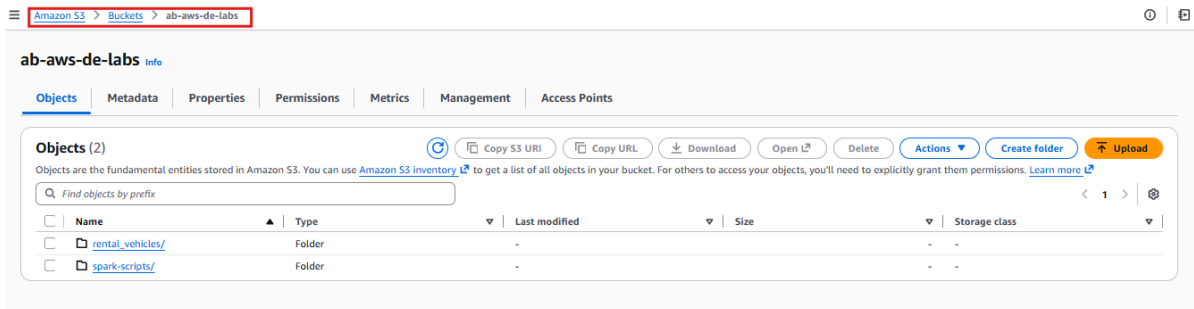
Each step is a unit of work that contains instructions to manipulate data for processing by software installed on the cluster.

Concurrent steps: 1 [Filter steps by status](#) [Find steps](#)

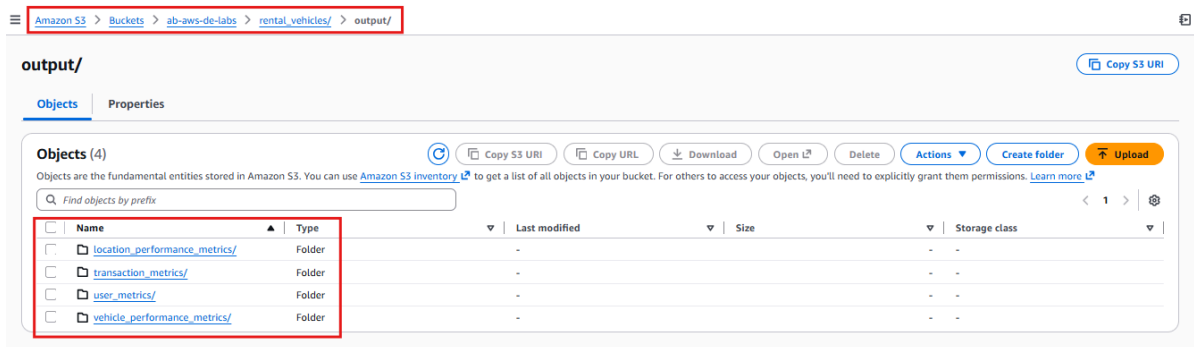
	Step ID	Status	Name	YARN Application ID	Log files	Creation time (UTC+05:30)	Start time
☐	s-060922830081HZKEUOX0	Completed	spark-job2	application_1770918557616_0004	No logs created yet	February 13, 2026 at 00:25	February 13, 2026 at 00:25
☐	s-05204231K365TNOVUWAH	Completed	spark-job1	application_1770918557616_0003	controller syslog stderr stdout	February 13, 2026 at 00:24	February 13, 2026 at 00:24

Verify Output in S3

- Go to S3 bucket



- Check output folder



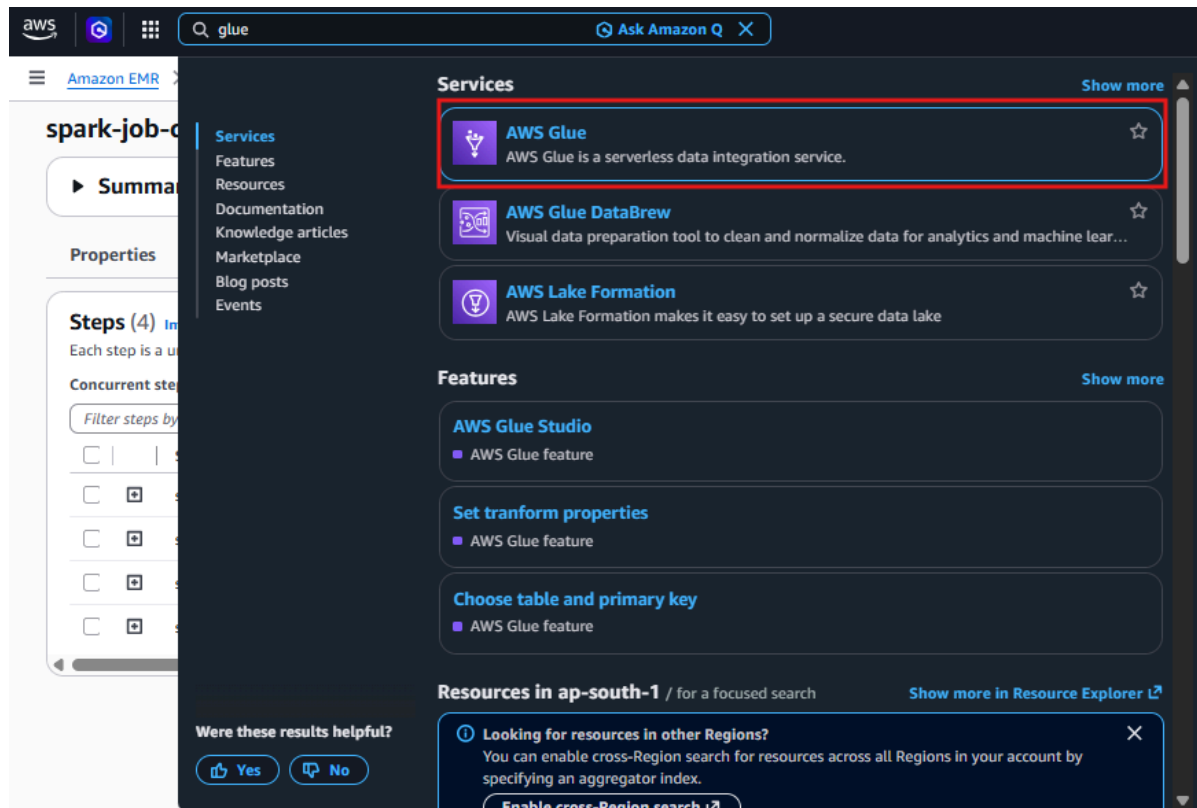
You should see:

- transaction_metrics
- user_metrics
- location_performance_metrics
- vehicle_performance_metrics

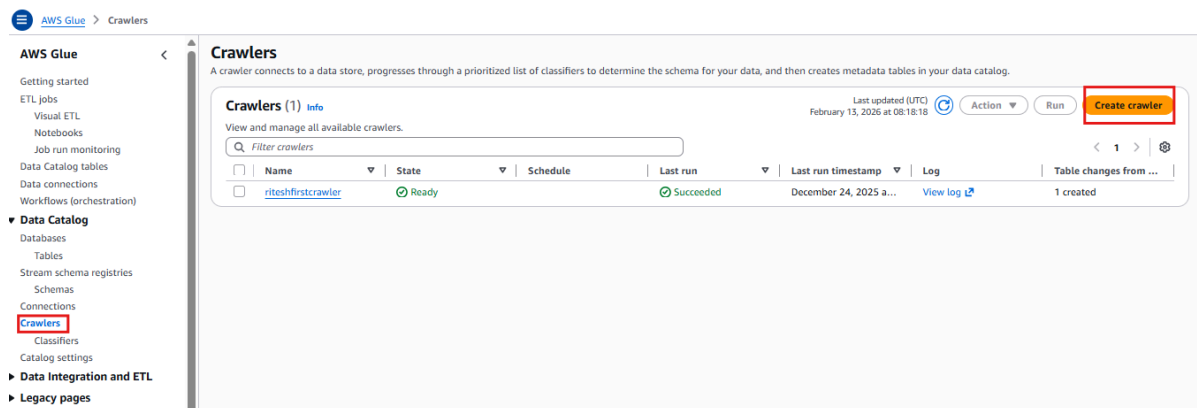
Part 4 – Create Glue Crawlers

(Repeat 4 times – one per dataset)

- Go to AWS Console → Glue



- Click Crawlers → Create Crawler



- Name the crawler

AWS Glue > Crawlers > Add crawler

AWS Glue

- Getting started
- ETL jobs
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 - Job run monitoring
- Data Catalog tables
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 - Classifiers
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What's New ⓘ
Documentation ⓘ
AWS Marketplace

Step 1
● **Set crawler properties**

Step 2
○ Choose data sources and classifiers

Step 3
○ Configure security settings

Step 4
○ Set output and scheduling

Step 5
○ Review and create

Set crawler properties

Crawler details ⓘ

Name

location_performance_metrics_crawler

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Enter a description

Descriptions can be up to 2048 characters long.

► **Tags - optional**

Use tags to organize and identify your resources.

Cancel **Next**

- Add data source → Browse S3

AWS Glue > Crawlers > Add crawler

AWS Glue

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What's New ⓘ
Documentation ⓘ
AWS Marketplace

Step 1
● Set crawler properties

Step 2
● **Choose data sources and classifiers**

Step 3
○ Configure security settings

Step 4
○ Set output and scheduling

Step 5
○ Review and create

Choose data sources and classifiers

Data source configuration

Is your data already mapped to Glue tables?

☒ Not yet
Select one or more data sources to be crawled.

☐ Yes
Select existing tables from your Glue Data Catalog.

Data sources (0) ⓘ

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
You don't have any data sources.		

Add a data source

► **Custom classifiers - optional**

A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.

Cancel Previous **Next**

×

Add data source

Data source
Choose the source of data to be crawled.

S3 ▾

Network connection - optional
Optionally include a Network connection to use with this S3 target. Note that each crawler is limited to one Network connection so any other S3 targets will also use the same connection (or none, if left blank).

▾

↺

Clear selection

Add new connection ↗

Location of S3 data

☒ In this account
☐ In a different account

S3 path
Browse for or enter an existing S3 path.

✕

View ↗

Browse S3

All folders and files contained in the S3 path are crawled. For example, type `s3://MyBucket/MyFolder/` to crawl all objects in MyFolder within MyBucket.

Subsequent crawler runs
This field is a global field that affects all S3 data sources.

☒ Crawl all sub-folders
Crawl all folders again with every subsequent crawl.
☐ Crawl new sub-folders only
Only Amazon S3 folders that were added since the last crawl will be crawled. If the schemas are compatible, new partitions will be added to existing tables.
☐ Crawl based on events
Rely on Amazon S3 events to control what folders to crawl.

☐ Sample only a subset of files
☐ Exclude files matching pattern

Cancel

Add an S3 data source

- Click Next
- Under configure security settings —→ Select IAM Role

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Step 1: Set crawler properties
Step 2: Choose data sources and classifiers
Step 3: **Configure security settings**
Step 4: Set output and scheduling
Step 5: Review and create

Configure security settings

IAM role [Info](#)

Existing IAM role: custom-glue-role [View](#)

[Create new IAM role](#) [Update chosen IAM role](#)

Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.

Lake Formation configuration - optional

Allow the crawler to use Lake Formation credentials for crawling the data source. [Learn more.](#)

☐ Use Lake Formation credentials for crawling S3 data source

Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.

Security configuration - optional

Enable at-rest encryption with a security configuration.

[Cancel](#) [Previous](#) [Next](#)

Attach Required IAM Permissions

To Glue Role attach:

AWS Glue

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Configure security settings

IAM role [Info](#)

Existing IAM role: custom-glue-role [View](#)

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Security configuration - optional

Enable at-rest encryption with a security configuration.

[Cancel](#) [Previous](#) [Next](#)

Identity and Access Management (IAM)

[Search IAM](#)

Dashboard

Access Management

- User groups
- Users
- Roles**
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

[New](#)

custom-glue-role [Info](#)

Allows Glue to call AWS services on your behalf.

[Delete](#)

[Edit](#)

Summary

Creation date
February 06, 2026, 14:58 (UTC+05:30)

ARN
[arn:aws:iam::463646775279:role/custom-glue-role](#)

Last activity
2 days ago

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies (3) [Info](#)

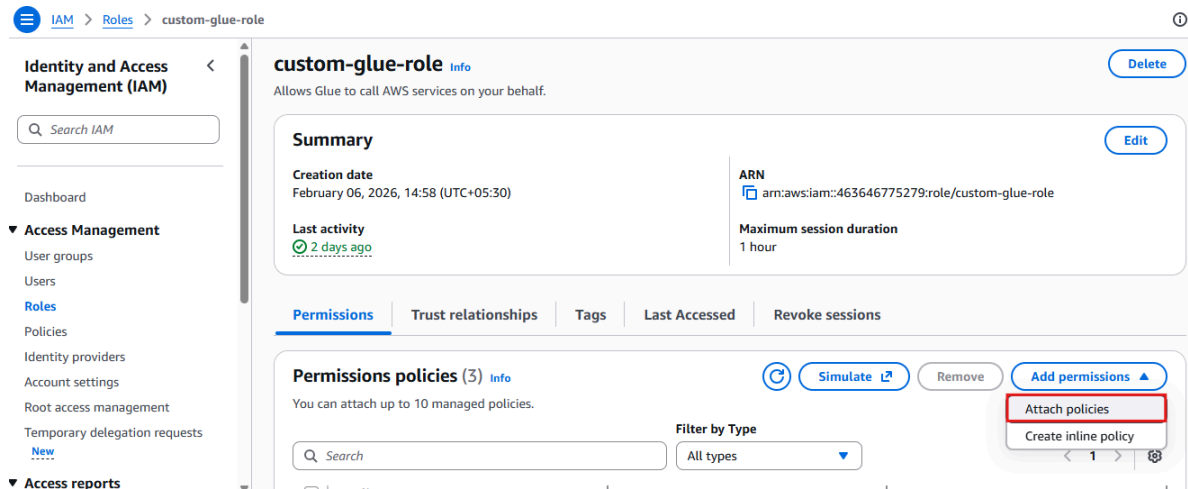
You can attach up to 10 managed policies.

[Simulate](#) [Remove](#) [Add permissions](#)

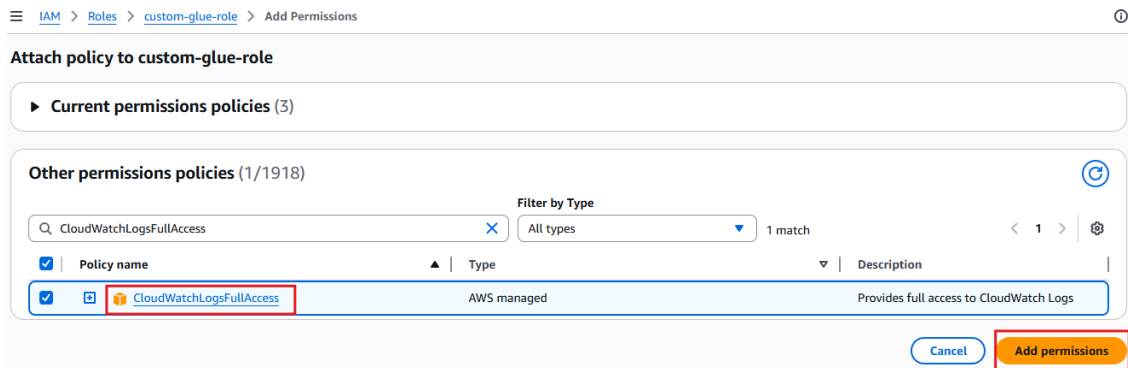
[Filter by Type](#)
All types

[1](#)

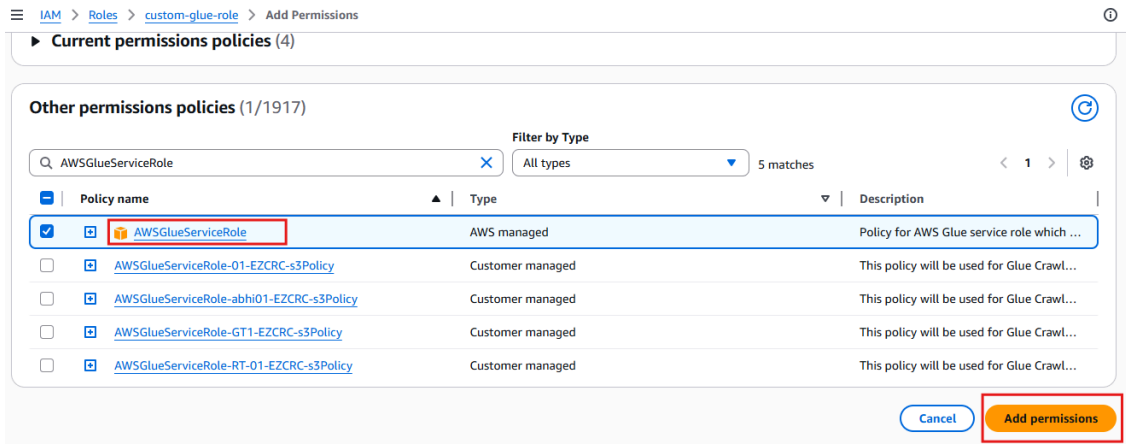
- Click Add permissions —→ Attach policies



- Add the following policies:
 - CloudWatchLogsFullAccess

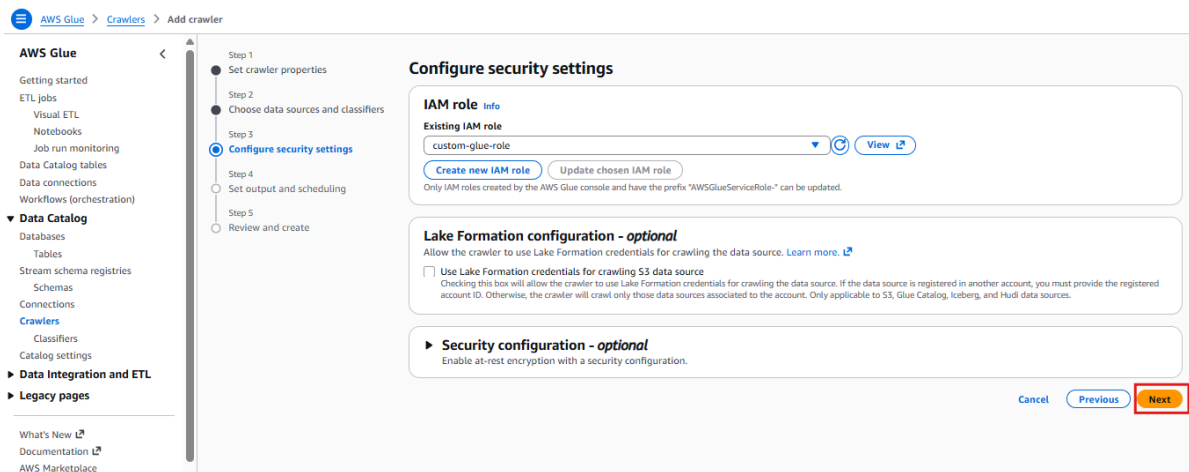


- AWSGlueServiceRole



Reason:

- Logs → CloudWatch
- Tables → Data Catalog
- Get back to AWS Glue —→ Click Next



- Under **Set output and scheduling** —→ Click on **Add database**

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Documentation [i](#)
AWS Marketplace

Step 1: Set crawler properties
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Step 5: Review and create

Set output and scheduling

Output configuration [info](#)

Target database
Choose a database ▼ [🔄](#)
Clear selection **Add database** [↗](#)

Table name prefix - optional
Type a prefix added to table names

Maximum table threshold - optional
This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If not set, the crawler will automatically generate the number of tables depending on the data schema.
Type a number greater than 0

Advanced options

Crawler schedule
You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like [cron](#) syntax. [Learn more](#) [↗](#).

Frequency
On demand ▼

Cancel Previous **Next**

Create Database

- Name database : rental_vehicles_db → Create Database

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AWS Marketplace

Create a database
Create a database in the AWS Glue Data Catalog.

Database details

Name
rental_vehicles_db
Database name is required, in lowercase characters, and no longer than 255 characters.

Description - optional
Enter text
Descriptions can be up to 2048 characters long.

Database settings

Location - optional
Set the URI location for use by clients of the Data Catalog.

Cancel **Create database**

- Get back to AWS Glue → Select database → Click next

Set output and scheduling

Output configuration [Info](#)

Target database
rental_vehicles_db

[Clear selection](#) [Add database](#)

Table name prefix - optional
Type a prefix added to table names

Maximum table threshold - optional
This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If not set, the crawler will automatically generate the number of tables depending on the data schema.
Type a number greater than 0

Advanced options

Crawler schedule
You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like cron syntax. [Learn more](#)

Frequency
On demand

[Cancel](#) [Previous](#) [Next](#)

- Click Create Crawler

Review and create

Set crawler properties

Name	Description	Tags
location_performance_metrics_crawler	-	-

Step 2: Choose data sources and classifiers [Edit](#)

Data sources (1) [Info](#)
The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
S3	s3://ab-aws-de-labs/rental_vehicles/output/lo...	Recrawl all

Step 3: Configure security settings [Edit](#)

Configure security settings

IAM role	Security configuration	Lake Formation configuration
custom-glue-role	-	-

Step 4: Set output and scheduling [Edit](#)

Set output and scheduling

Database	Table prefix - optional	Maximum table threshold - optional	Schedule
rental_vehicles_db	-	-	On demand

[Cancel](#) [Previous](#) [Create crawler](#)

- Click Run Crawler

location_performance_metrics_crawler

Last updated (UTC)
February 13, 2026 at 09:16:51

[Run crawler](#) [Edit](#) [Delete](#)

Crawler properties

Name	IAM role	Database	State
location_performance_metrics_crawler	custom-glue-role	rental_vehicles_db	READY

Description	Security configuration	Lake Formation configuration	Table prefix
-	-	-	-

Maximum table threshold
-

[Advanced settings](#)

- Wait until status shows completed

The screenshot shows the AWS Glue console for the crawler 'location_performance_metrics_crawler'. The crawler properties are as follows:

Property	Value
Name	location_performance_metrics_crawler
IAM role	custom-glue-role
Database	rental_vehicles_db
State	READY
Description	-
Security configuration	-
Lake Formation configuration	-
Table prefix	-
Maximum table threshold	-

The crawler runs section shows one completed run:

Start time (UTC)	End time (UTC)	Current/last duration	Status	DPU hours	Table changes
February 13, 2026 at 09:18:20	February 13, 2026 at 09:19:00	40 s	Completed	0.115	1 table changes, 0 partition changes

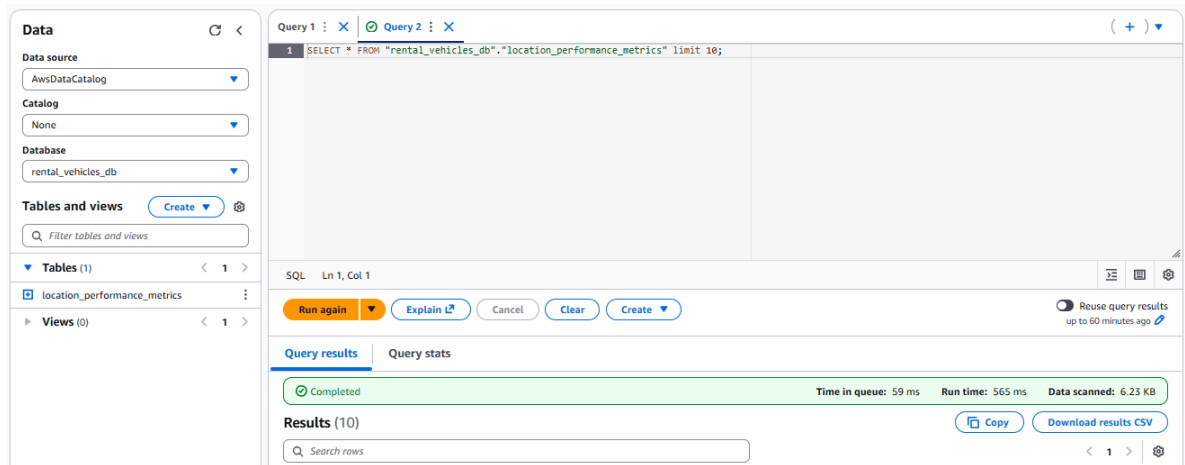
Note: Similarly create the remaining 3 crawlers — follow same steps

Verify Table in Athena

- Go to Athena —→ Select database: rental_vehicles_db

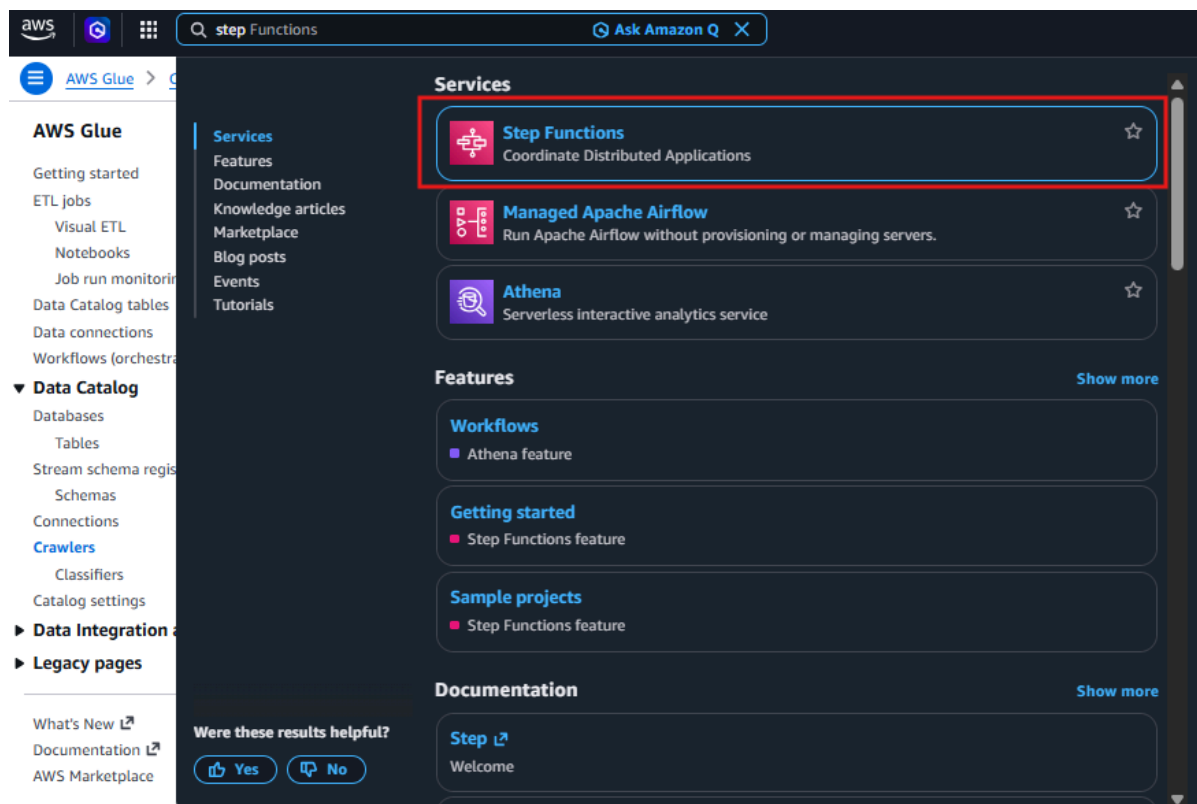
The screenshot shows the Amazon Athena Query Editor. The 'Database' dropdown menu is selected and shows 'rental_vehicles_db' as the chosen option. The 'Data' section on the left shows the 'Data source' as 'AwsDataCatalog' and the 'Catalog' as 'None'. The 'Tables and views' section shows a list of tables, including 'location_performance_metrics'.

- Preview table



Part 5 – Automate Using Step Functions

- Go to AWS search bar —→ Click Step Functions



- Click Create State Machine

Step Functions > State machines

State machines (2) View execution counts View details Edit Copy to new Delete **Create state machine**

Search for state machines Any type

Name	Type	Creation date	Status
MyStateMachine-plogk5i4n	Standard	Aug 25, 2024, 16:53:45 (UTC+05:30)	Active
MyStateMachine-onlyvly364	Standard	Aug 25, 2024, 16:43:10 (UTC+05:30)	Active

Developer resources: Online learning workshop, Local Development, Data flow simulator, Feature spotlight, Documentation

Join our feedback panel

- Select Blank → Code

Create state machine

Create from blank Create your own workflow from scratch.

Create from template Choose a workflow template that matches your use case.

State machine name
State machine name can be edited in the Config section of Workflow Studio before creating your state machine. After creation, the name cannot be changed.
 Generate

Must be 1-80 characters. Can use alphanumeric characters, dashes, and underscores.

State machine type
State machine type can be edited in the Config section of Workflow Studio before creating your state machine. After creation, the type cannot be changed.

☒ **Standard**
Durable workflows for ETL, ML, e-commerce and automation. Standard workflows can run for up to 1 year, and history is stored in Step Functions for auditing and playback. Supported by a feature-rich console debugger. Recommended for new users. [Learn more](#)

☐ **Express**
Low cost, high scale workflows for streaming data processing and microservice APIs. Express workflows can run for up to 5 minutes, and history can be streamed to CloudWatch Logs. [Learn more](#)

Exit **Continue**

- Click on code tab → Paste Step Functions code

emr-spark-job-execution Design **Code** Config

Workflow not created Cancel Actions Create

Undo Redo Format Copy Commands View docs

```

1
2
3
4
5
6
7
8
9
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11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29

```

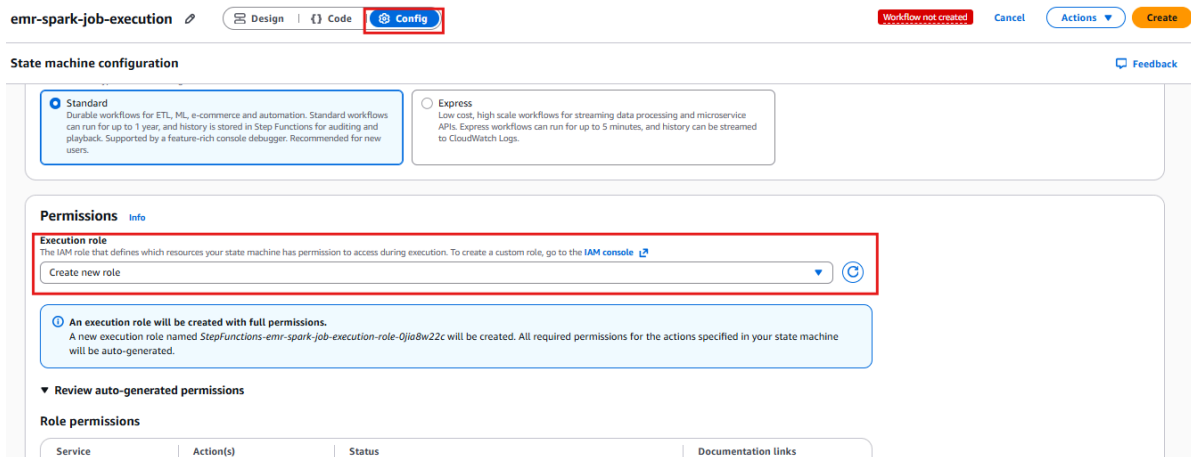
```

{
  "Comment": "Create an EMR Cluster, run Spark jobs, and terminate the cluster regard",
  "StartAt": "CreateEMRCluster",
  "States": {
    "CreateEMRCluster": {
      "Type": "Task",
      "Resource": "arn:aws:states:::elasticmapreduce:createCluster.sync",
      "Parameters": {
        "Name": "EMR Cluster with Specific Applications",
        "ReleaseLabel": "emr-7.1.0",
        "Applications": [
          {
            "Name": "Spark"
          }
        ],
        "Instances": {
          "InstanceGroups": [
            {
              "Name": "Master nodes",
              "InstanceRole": "MASTER",
              "InstanceType": "m5.xlarge",
              "InstanceCount": 1,
              "Market": "ON_DEMAND"
            },
            {
              "Name": "Core nodes",
              "InstanceRole": "CORE",
              "InstanceType": "m5.xlarge",
              "InstanceCount": 2,

```

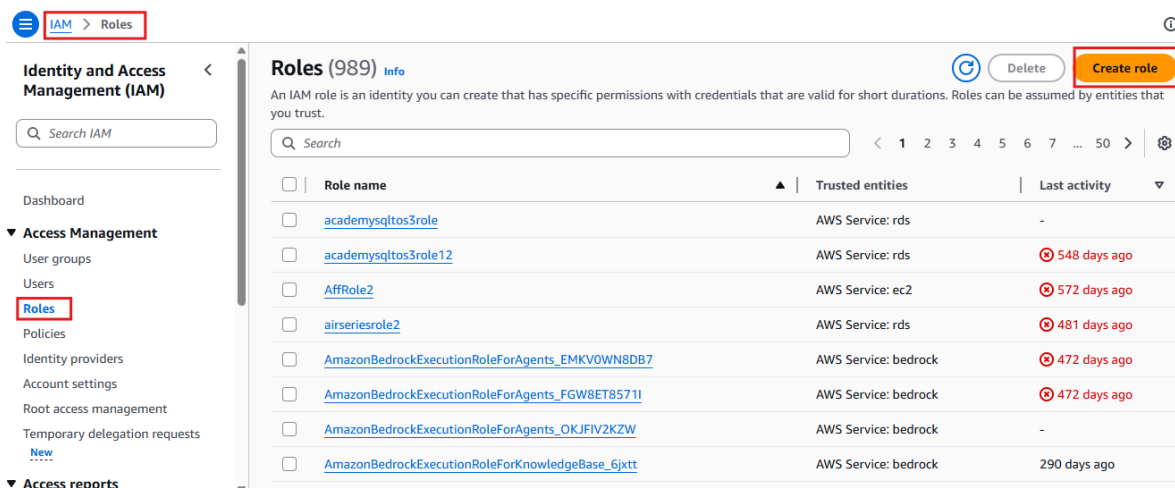
Diagram: Start → EMR: CreateCluster CreateEMRCluster → EMR: AddStep AddSparkJob1 → Catch #1 → EMR: AddStep AddSparkJob2 → Catch #1 → EMR: TerminateCluster

- Click on config tab → come to Execution Role



Create Step Function Execution Role

- IAM → Roles —→ Create Role



- Select use case —→ Step Functions

IAM > Roles > Create role

☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
Step Functions

Choose a use case for the specified service.
Use case
☒ Step Functions
Allows Step Functions to access AWS resources on your behalf.

Cancel **Next**

- Give execution role a name —→ Click create role

IAM > Roles > Create role

Step 1
● Select trusted entity
Step 2
● Add permissions
Step 3
● **Name, review, and create**

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.
custom-steps-function-execution-role
Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

Description
Add a short explanation for this role.
Allows Step Functions to access AWS resources on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-./[()!#\$%^&*(){};:'"

Step 1: Select trusted entities **Edit**

Trust policy

- Click on Add permissions

IAM > Roles > custom-steps-function-execution-role

Identity and Access Management (IAM)
Search IAM

Dashboard

▼ **Access Management**
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests
New

▼ **Access reports**

Summary **Edit**

Creation date
February 13, 2026, 16:04 (UTC+05:30)

ARN
arn:aws:iam::463646775279:role/custom-steps-function-execution-role

Last activity
-

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Last Accessed | Revoke sessions

Permissions policies **Info** **Simulate** **Remove** **Add permissions**

You can attach up to 10 managed policies.

Filter by Type
All types

Attach policies
Create inline policy

Policy name | Type | Attached entities

Loading policies

Attach:

- Inline policy (EMR execution JSON)

Step 1
● Specify permissions
○ Step 2
○ Review and create

Specify permissions Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor
Visual **JSON** Actions 

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": "iam:PassRole",  
7       "Resource": [  
8         "arn:aws:iam:463646775279:role/emr-custom-service-role",  
9         "arn:aws:iam:463646775279:role/emr-ec2-custom-role"  
10      ],  
11     "Condition": {  
12       "StringEquals": {  
13         "iam:PassedToService": [  
14           "elasticmapreduce.amazonaws.com",  
15           "ec2.amazonaws.com"  
16         ]  
17     }  
18   }  
19 ]  
20 }
```

Edit statement

Select a statement
Select an existing statement in the policy or add a new statement.
[+ Add new statement](#)

Step 1
● Specify permissions
○ Step 2
● Review and create

Policy details

Policy name
Enter a meaningful name to identify this policy.

Maximum 128 characters. Use alphanumeric and '+', '@', '_', '-' characters.

Permissions defined in this policy Info [Edit](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Allow (2 of 462 services) [Show remaining 460 services](#)

Service	Access level	Resource	Request condition
EMR	Limited: List, Read, Write	All resources	None
IAM	Limited: Write	Multiple	iam:PassedToService = elasticmapreduce.amazonaws.co

- CloudWatchLogsFullAccess
- S3FullAccess

Identity and Access Management (IAM)

Search IAM

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Access Management

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- Policies
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- Temporary delegation requests

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (4) Info

You can attach up to 10 managed policies.

Filter by Type

Search

All types

Policy name	Type	Attached entities
AmazonS3FullAccess	AWS managed	91
AWSLambdaRole	AWS managed	4
CloudWatchLogsFullAccess	AWS managed	7
custom-policy-emr-step-functions	Customer inline	0

Permissions boundary (not set)

- Get back to step functions tab → Select this role in Step Function
- Set log level to ALL

emr-spark-job-execution

Design | Code | **Config**

Workflow not created

Cancel

Actions

Create

State machine configuration

Feedback

Permissions Info

Execution role

The IAM role that defines which resources your state machine has permission to access during execution. To create a custom role, go to the IAM console.

custom-steps-function-execution-role

View in IAM

Logging Info

You can log your state machine's execution history to CloudWatch Logs. For Express state machines, you must enable logging to inspect and debug executions. CloudWatch Logs charges apply. Learn more.

Log level

Indicates which execution history events to log

ALL

Include execution data

Log execution input, data passed between states, and execution output

CloudWatch log group

Create new log group

/aws/vendedlogs/states/emr-spark-job-execution-Logs

- Click Create

Start Execution

- Click on Execute

emr-spark-job-execution Standard Design Code Config Exit Actions **Execute** Save

Undo Redo Zoom in Zoom out Center Duplicate Delete Feedback

Search

ACTIONS Flow Patterns Info

MOST POPULAR

- AWS Lambda Invoke
- Amazon SNS Publish
- Amazon ECS RunTask
- AWS Step Functions StartExecution
- AWS Glue StartJobRun

HTTP APIS

- HTTP Endpoint Call HTTPS APIs

Workflow Definition

The top level state machine properties for this workflow. [Learn more](#)

State machine query language | Info
JSONPath

Start at
The state that is the starting point of the workflow.
CreateEMRCluster

Comment - optional
A human-readable description of the state machine.
Create an EMR Cluster, run Spark jobs, and terminate the cluster regardless of job success

TimeoutSeconds - optional
The maximum number of seconds an execution of the state machine can run. If it runs longer than the specified time, the execution fails with a `States.Timeout`.
600

Start execution

Name
1a41fd6c-d721-4eca-b00c-b03d992a5944
Must be 1-80 characters. Can use alphanumeric characters, dashes, or underscores.

Input - optional
Enter input values for this execution in JSON format.
[Format JSON](#) [Export](#) [Import](#)

```
1 {  
  "Comment": "Insert your JSON here"  
}
```

[Start execution with latest revision](#)

☐ Open in a new browser tab

[Cancel](#) **Start execution**

- Monitor execution

It will:

1. Create EMR Cluster
 2. Run Spark Job 1
 3. Run Spark Job 2
 4. Terminate Cluster
- Verify new EMR cluster is created
 - Wait until execution succeeds

Step Functions > State machines > emr-spark-job-execution > Execution: 1a41fd6c-d721-4eca-b00c-b03d992a5944

Execution: 1a41fd6c-d721-4eca-b00c-b03d992a5944

Details Execution input and output Definition

Execution status: **Succeeded**

Execution type: Standard

Execution ARN: [arn:aws:states:ap-south-1:463646775279:execution:emr-spark-job-execution:1a41fd6c-d721-4eca-b00c-b03d992a5944](#)

IAM role ARN: [arn:aws:iam::463646775279:role/custom-steps-function-execution-role](#)

State transitions: [Learn more](#)

6

Execution Logs: [Learn more](#)

[CloudWatch Logs](#)

Start time: Feb 13, 2026, 17:04:19.352 (UTC+05:30)

End time: Feb 13, 2026, 17:14:09.235 (UTC+05:30)

Duration: 00:09:49.883

Alias: -

Version: -

Graph view Table view

Graph view

Start

Step details

Choose a step to view its details.

Graph view

Start

Amazon EMR Service Integ: CreateEMRCluster

Amazon EMR Service Integ: AddSparkJob1

Amazon EMR Service Integ: AddSparkJob2

Amazon EMR Service Integ: TerminateCluster

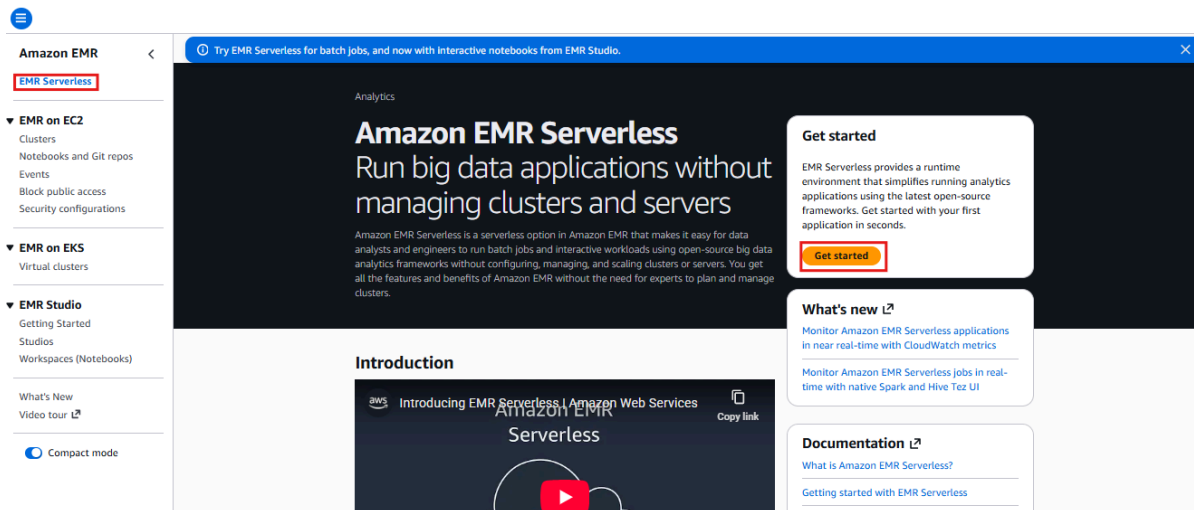
End

In progress Failed Caught error Canceled Succeeded

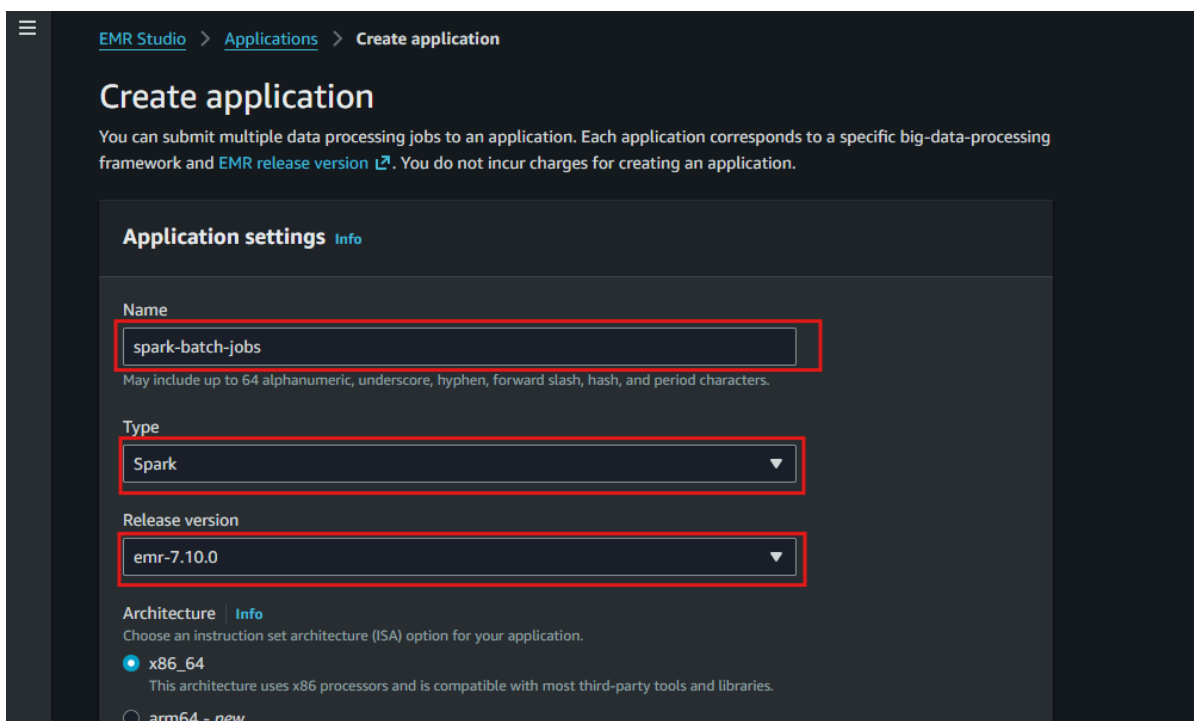
Event view State view

Part 6 – Run on EMR Serverless

- Go to EMR → EMR Serverless —→ Click Get Started



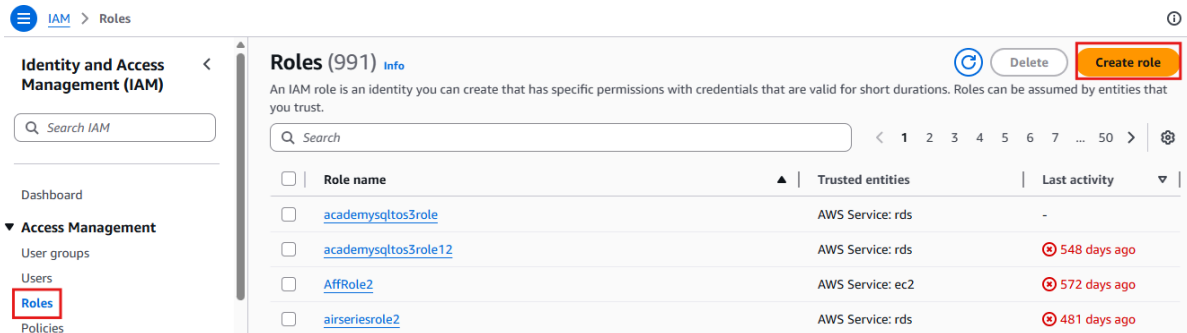
- Create Application
- Name: spark-batch-jobs
- Type: Spark
- Version: 7.1.0



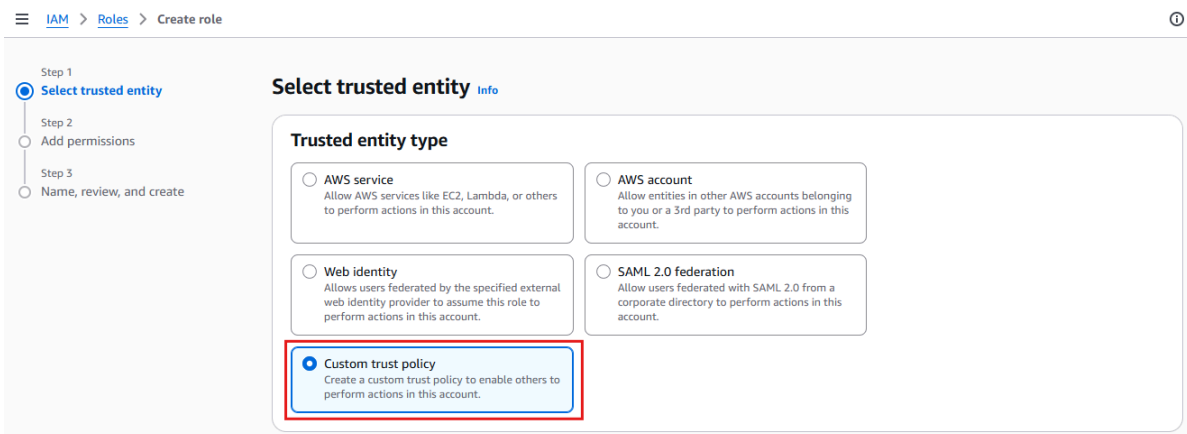
- Click create and start application

Create Runtime IAM Role

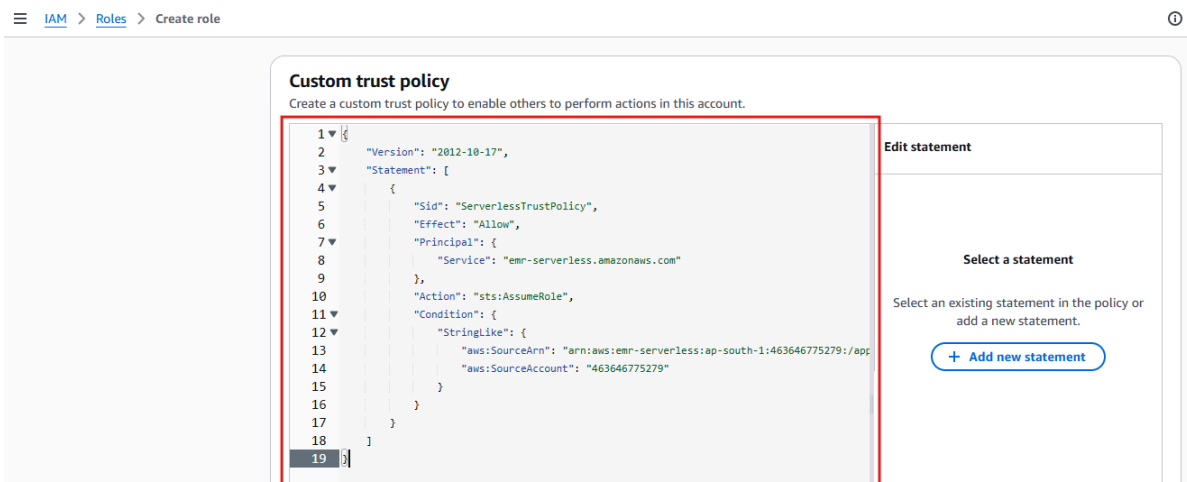
- IAM → Create Role



- Select Custom Trust Policy



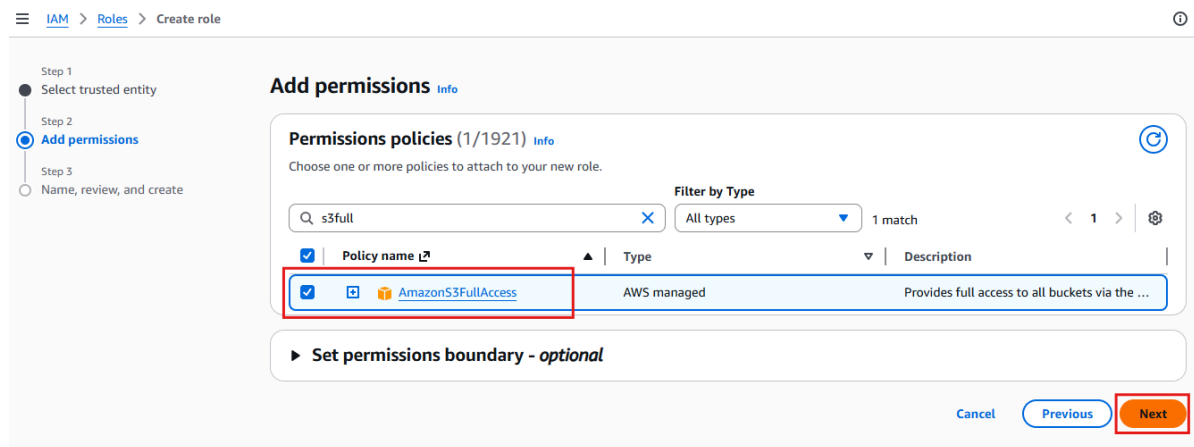
- Paste EMR Serverless trust JSON



- Update:
- AWS Account ID
- Application ID
- Region
- Click Next

Attach:

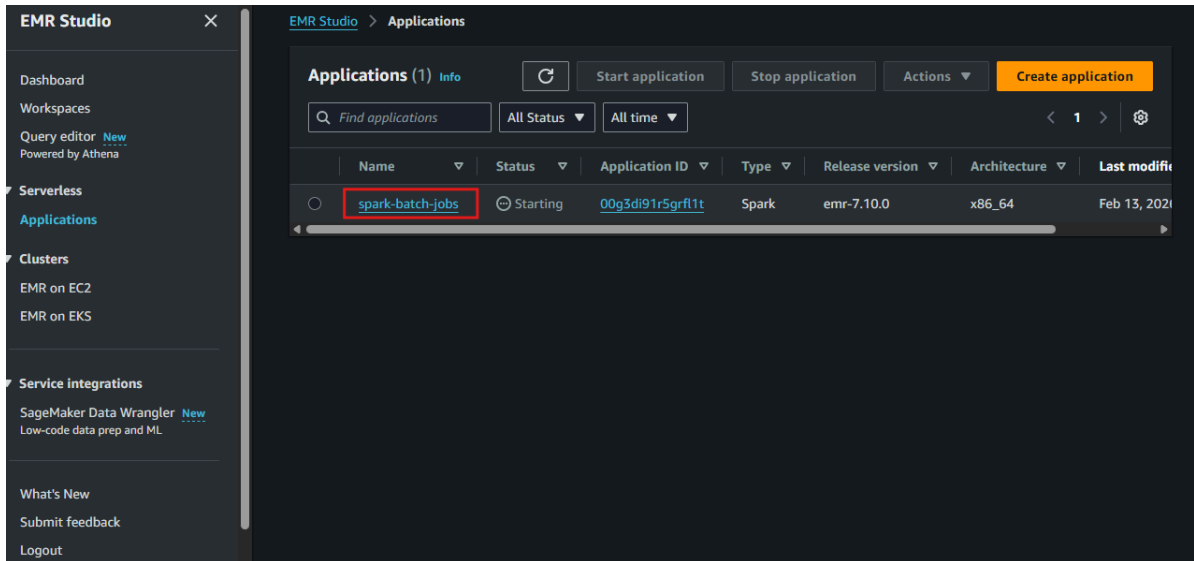
- S3 Full Access



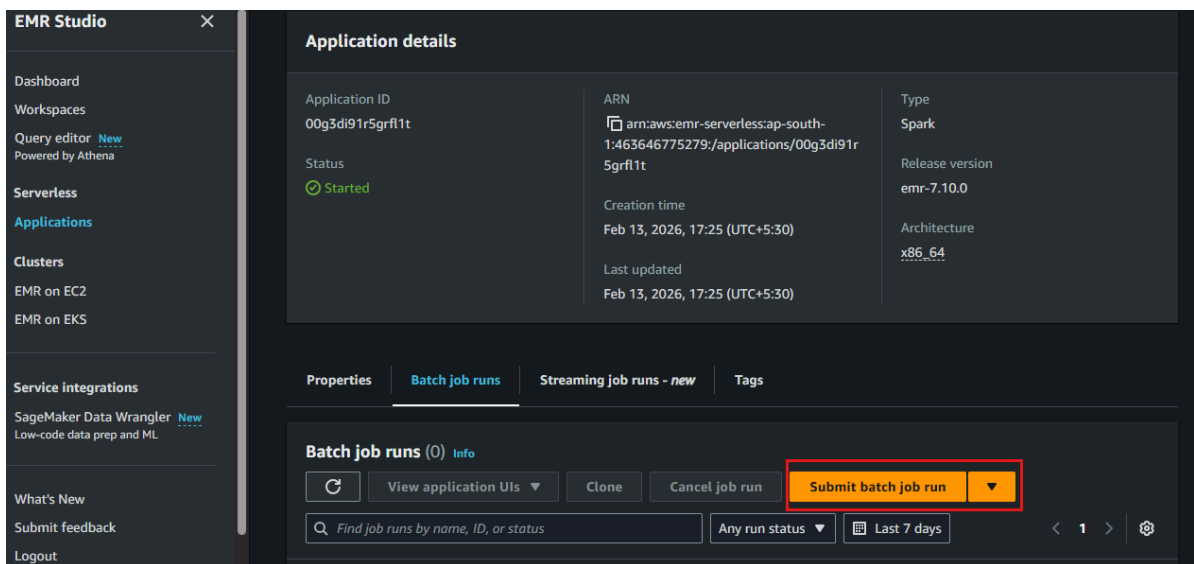
- Create Role

Submit Serverless Spark Job

- Open EMR Studio



- Click Submit Batch Job



- Name job —→ Select runtime role —→ Provide S3 script location

EMR Studio > Applications > spark-batch-jobs > Submit job run

Submit batch job run

Job details Info

Name

Runtime role
The IAM role assumed by the job. This role must have permissions to access your data sources, targets, scripts, and any libraries used by the job. [Learn more](#)

☐ Specify an additional inline Runtime IAM policy

Script location
The location of the main JAR or Python script in Amazon S3 that you want to run

Script arguments
An array of arguments passed to your main JAR or Python script. Your code should handle reading these parameters. Each argument in the array must be separated by a comma.

- Click Submit job run
- Wait until status shows Success

EMR Studio

- Dashboard
- Workspaces
- Query editor [New](#)
Powered by Athena
- ▼ Serverless
 - Applications**
- ▼ Clusters
 - EMR on EC2
 - EMR on EKS
- ▼ Service integrations
 - SageMaker Data Wrangler [New](#)
Low-code data prep and ML
- What's New
- Submit feedback
- Logout

Status: Started

1:463646775279:/applications/00g3di91r5grfl1t

Release version: **emr-7.10.0**

Creation time: Feb 13, 2026, 17:25 (UTC+5:30)

Architecture: **x86_64**

Last updated: Feb 13, 2026, 17:47 (UTC+5:30)

Properties | **Batch job runs** | Streaming job runs - new | Tags

Batch job runs (1) Info

Find job runs by name, ID, or status

Any run status | Last 7 days

☐	Job run name	Run status	Job run ID	Retry attempts	Driver log files	Start time (UTC+5:30)
☐	first-Spark-Job	Success	00g3dil9v81co01v	1	View logs	Feb 13, 2026, 17:47

Final Outcome of Lab

You successfully:

- Processed big data using Spark

- Ran jobs on EMR (Provisioned)
- Automated workflow using Step Functions
- Queried data using Athena
- Implemented EMR Serverless
- Built a complete Data Lake pipeline